

Problem B. Grid

Time Limit 1000 ms

Mem Limit 131072 kB

OS Linux

There is a $n \times n$ grid D where each cell contains either 1 or 0.

Your task is to create a program that takes the grid data as input and computes the greatest number of consecutive 1s in either vertical, horizontal, or diagonal direction.

For example, the consecutive 1s with greatest number in the figure below is circled by the dashed-line.

0	0	0	1	1
0	0	1	0	1
0	1	0	0	0
1	0	1	0	1
0	0	0	1	0

The size of the grid n is an integer where $2 \leq n \leq 255$.

Input

The input is a sequence of datasets. The end of the input is indicated by a line containing one zero. Each dataset is formatted as follows:

```
n
D11 D12 ... D1n
D21 D22 ... D2n
.
.
Dn1 Dn2 ... Dnn
```

Output

For each dataset, print the greatest number of consecutive 1s.

Input	Output
5	4
00011	8
00101	1
01000	0
10101	
00010	
8	
11000001	
10110111	
01100111	
01111010	
11111111	
01011010	
10100010	
10000001	
2	
01	
00	
3	
000	
000	
000	
0	